

ARXIV:2405.04434

DEEPSEEK-V2

A Strong, Economical, and Efficient
Mixture-of-Experts Language Model

236B

TOTAL PARAMS

21B

ACTIVATED

128K

CONTEXT

Pretrained on 8.1T tokens · Multi-Head Latent Attention + DeepSeekMoE

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed



Training Costs

Larger models yield better intelligence but training costs grow prohibitively with scale



KV Cache Bottleneck

Standard MHA stores full key-value pairs per head per token, creating memory and speed bottlenecks

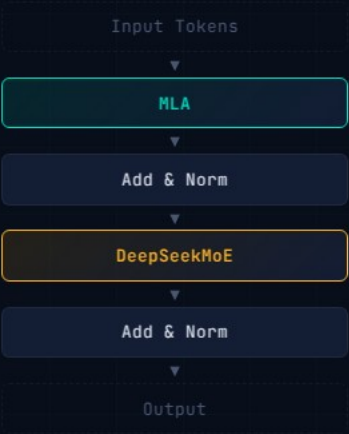


Breaking the Tradeoff

MoE architectures + efficient attention can deliver top-tier performance at a fraction of the cost

DeepSeek-V2 Architecture

MLA + DeepSeekMoE in a Transformer Block



Multi-Head Latent Attention

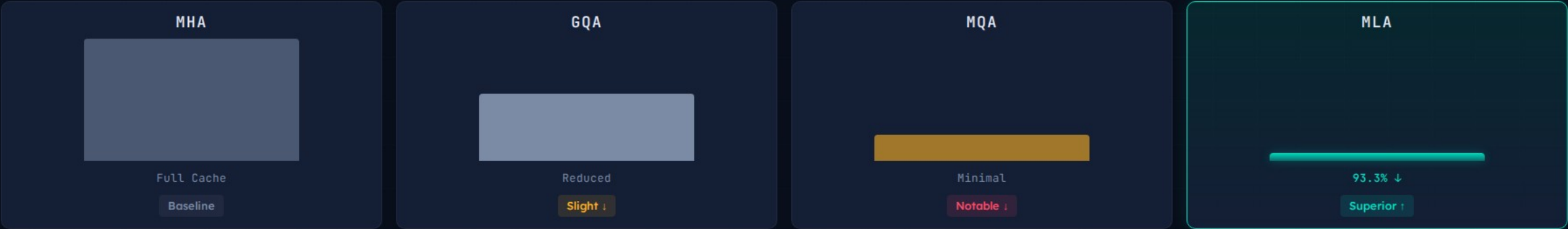
Low-rank key-value joint compression into compact latent vectors. Full keys and values reconstructed on-the-fly via learned projections with decoupled RoPE positional encoding.

DeepSeekMoE FFN

Fine-grained mixture of experts with shared experts (always active) and routed experts (top-K sparse activation). Only 21B of 236B total parameters activated per token.

Multi-Head Latent Attention

Compresses KV Cache by 93.3% While Improving Performance



Low-rank KV joint compression with decoupled RoPE — better performance than MHA with dramatically less cache

DeepSeekMoE: Fine-Grained Expert Routing

Shared Expert Isolation + Sparse Routed Activation

Shared Experts (N_s) – Always Active



Capture common knowledge across all tokens

Routed Experts (N_r) – Top- K_r Selected



Fine-grained specialization via sparse routing

Device-Limited Routing

Each token routed to experts on at most M devices, reducing cross-node communication overhead

Load Balancing

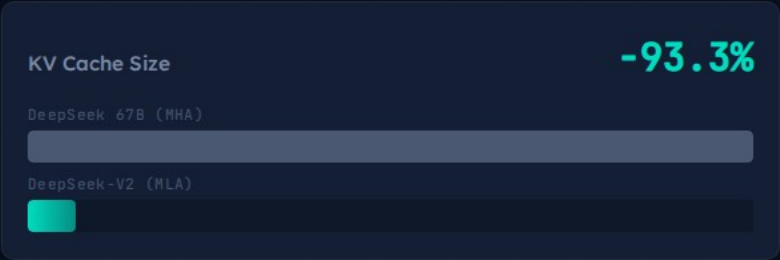
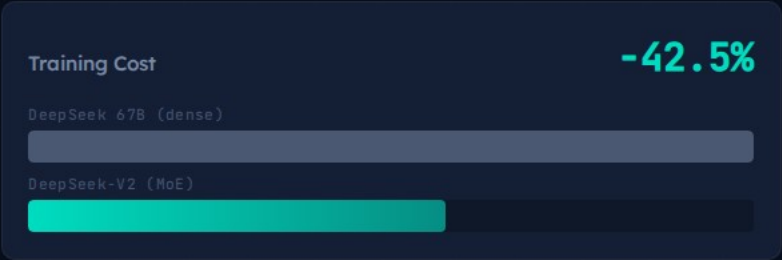
Auxiliary losses enforce balanced expert and device utilization; token-dropping during training prevents overflow

236B → 21B Activated

Massive total capacity with economical per-token compute, enabling training at 42.5% lower cost than a 67B dense model

Efficiency Gains vs. DeepSeek 67B

42.5% Training Cost Reduction · 93.3% KV Cache Reduction · 5.76× Throughput



MMLU vs. Activated Parameters

DeepSeek-V2 Achieves Pareto-Optimal Efficiency Among Open-Source Models



Alignment & Chat Evaluation

DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

Post-Training Pipeline

- 1

Supervised Fine-Tuning
1.5M conversational sessions
- ▼
- 2

GRPO Reinforcement Learning
Group Relative Policy Optimization

AlpacaEval 2.0 38.9% Length-controlled win rate

MT-Bench 8.97 Overall score

AlignBench 7.91 Chinese · Outperforms all open-source & most closed-source models

Key Takeaways

Architectural Innovation Unlocks Efficient Scaling

- 01 **MLA surpasses MHA** while reducing KV cache by 93.3% — a strictly better alternative to GQA and MQA for inference efficiency
- 02 **DeepSeekMoE with fine-grained experts** enables training a 236B-parameter model at 42.5% lower cost than a 67B dense model
- 03 **Only 21B activated parameters** needed to match or exceed 70B-class dense models — DeepSeek-V2 scores 78.5 on MMLU vs. 77.0 for LLaMA 3 70B
- 04 **5.76× throughput improvement** over DeepSeek 67B makes DeepSeek-V2 practical for large-scale deployment
- 05 **Strongest open-source Chinese chat model** and competitive with closed-source systems in English open-ended evaluation